# Sahaj Bhadja

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## Summary

Software engineer with 2+ years building **enterprise-scale AI products, customer data platforms, real-time workflows, and personalized experiences** using Java, Python, React, PostgreSQL, Redis, AWS, Docker, Kubernetes, and distributed systems.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*

## Experience

**USC Dornsife** **LA, California**
*Software Engineer* *September 2025 – Present*

- Architected multi-tenant customer data infrastructure with PostgreSQL, SQL Server, and row-level isolation, supporting 500+ users and 10,000+ records while reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant distributed workflows using FastAPI workers, explicit states, and failure isolation, processing 5,000+ monthly jobs while increasing concurrent throughput 3x and sustaining 99%+ production reliability at scale.

**Grad8** **LA, California**
*Software Engineer Intern* *June 2025 – August 2025*

- Shipped an AI-powered React and Python platform serving 150+ instructors and 10,000+ daily requests, reducing feedback turnaround 60% through scalable APIs, concurrent execution, and product workflow improvements.
- Integrated OpenAI, Anthropic, and Qwen behind a switchable model layer, adding fallbacks, token-cost tracking, and monitoring to maintain consistently reliable outputs during high-volume production workloads at scale.

**USC AD Labs** **Los Angeles, CA**
*ML Research Assistant* *January 2025 – May 2025*

- Developed Python and PyTorch data pipelines processing 100,000+ autonomous-driving trajectory records, improving long-horizon planning 18% through feature engineering, attention-based modeling, iterative evaluation, and performance optimization at scale.
- Designed distributed representations across ego, road, and nearby-agent context signals, improving closed-loop stability 15% while maintaining experiment tracking, reproducible configurations, and technical documentation for cross-functional review at scale.

**Aasma Technology Solutions** **India**
*Software Developer Engineer - (SecureSync)* *August 2023 – July 2024*

- Developed Java and Python microservices with AWS Glue and S3 pipelines, consolidating JSON, CSV, and Parquet data for analytics while improving response times 30% and pipeline efficiency 68%.
- Maintained automated CI/CD and testing across distributed Linux, Windows, and macOS environments, executing 20+ test suites while collaborating with Microsoft and GitHub stakeholders through Agile releases at scale.

**Aasma Technology Solutions** **India**
*Full Stack Developer Intern - (VibrantChem)* *May 2023 – August 2023*

- Built eight customer-engagement microservices for profiles, interactions, billing, leads, preferences, and notifications using React, Node.js, WebSockets, and REST APIs, synchronizing two million records with 33% fewer conflicts.
- Converted analytics and monitoring signals into product improvements, reducing resource waste 42% and increasing lead conversions 15% while delivering automated tests, Docker artifacts, and staged cloud deployments.

## Skills

**Languages**: Java, Python, TypeScript, JavaScript, SQL, C++, Shell, Bash
**Applications**: React, Next.js, Node.js, Express.js, FastAPI, Flask, REST APIs, WebSockets
**Data**: PostgreSQL, MySQL, SQL Server, MongoDB, Redis, ETL pipelines, data modeling, analytics
**Infrastructure**: AWS, Azure, Docker, Kubernetes, Linux, distributed systems, CI/CD, Jenkins
**Product**: AI systems, personalization, customer engagement, monitoring, testing, Agile, performance optimization

## Projects

**Big-Brain** Live | *AI Retrieval, Customer Data*

- Built an LLaMA-2 interaction platform over 50,000+ records with semantic search, conversational workflows, secure access, and real-time collaboration, reducing analysis time 60% through scalable low-latency retrieval at scale.

**FinNexus** Live | *Real-Time Engagement, Enterprise Scale*

- Developed a TypeScript platform supporting 50,000+ users and one million transactions, delivering real-time WebSocket updates, advanced filtering, secure APIs, scalable MongoDB access, and Kubernetes-based deployment automation at scale.

**Trend Tracker with Fashion Recommendation System** Live | *Personalization*

- Built a React recommendation experience using CNN-based personalization, improving measured recommendation accuracy from 78% to 92% while serving three million users and generating 200,000 positive retentions overall.